Key Statistical and Mathematical Prerequisites

Yihao Yuan*

How to Use This Document

This document is designed to review previously acquired statistics knowledge. The purpose of this document is to review and introduce notation consistent with our course.

Section 1 is **required** and must be **mastered and committed to memory**. There will be no time to review this material in lectures. The remaining sections are strongly recommended. They review the basic statistical concepts. Reading them will greatly enhance your understanding of the course materials. If you aren't familiar with the basic concepts, please consult your basic stats book.

1 Mathematical Operations

Exponential Functions

A function having a constant base and a variable exponent is an exponential function. For example, $y = f(x) = b^x$ where b is the base and x is the exponent. Some rules of exponents:

- $b^u b^v = b^{u+v}$
- $(b^u)^v = b^{uv}$
- $(abc)^u = a^u b^u c^u$
- $(a/b)^u = a^u / b^u$
- $b^{-u} = 1/b^u$
- $b^u / b^v = b^{u-v}$
- $b^{1/v} = \sqrt[v]{b}$
- $b^0 = 1$ if $b \neq 0$

A frequently used exponential function is $y = e^x$ where $e = 2.71828\ldots$

Logarithms

By definition, the logarithm of a positive number y to a positive base b (other than $b = 1$), is the exponent x to which the base must be raised to obtain the number. Thus, if $y = b^x$ (and $b > 0$ but $b \neq 1$) then x is the logarithm of y to the base b. Therefore,

$$x = f(y) = \log_b(y)$$

is a logarithmic function. Note that exponential and logarithmic functions are inverse functions. Some rules of logarithms:

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- $\log_b(uv) = \log_b(u) + \log_b(v)$
- $\log_b(u/v) = \log_b(u) - \log_b(v)$
- $\log_b(u^n) = n\log_b(u)$
- $\log_b(b) = 1$
- $\log_b(1) = 0$

Two frequently used bases in logarithms are $b = 10$ ("common logarithms") and $b = e$ ("natural logarithms"). Note that "ln" is used as shorthand for $\log_e$.

Summation Notation

The summation notation, $\sum$, is a shorthand notation for summing over a group of terms. Thus, for $i = 1, 2, \ldots, N$:

$$\sum_{i=1}^{N} cX_i = cX_1 + cX_2 + \ldots + cX_N$$

where i is the summation index. Note that sum is a linear operator. Hence, for scalar c, we have

- $\sum cX_i = c\sum X_i$
- $\sum c = c\sum 1 = cN$
- $\sum (x_i - y_i) = \sum x_i - \sum y_i$

Summation notation is frequently used in statistics courses. Two examples are the sample mean and sample variance:

- Sample Mean: $\bar{Y} = \sum_{i=1}^{N} Y_i/N$
- Sample Variance: $s^2 = \sum_{i=1}^{N} \left(Y_i - \bar{Y}\right)^2/(N-1)$

One interesting property of the sample average is that the sum of deviations from the sample average is zero, i.e.,

$$\sum_{i=1}^{N} (Y_i - \bar{Y}) = 0$$

Proof:

$$\sum_{i=1}^{N} (Y_i - \bar{Y}) = \sum_{i=1}^{N} Y_i - \sum_{i=1}^{N} \bar{Y} = \frac{N}{N}\sum_{i=1}^{N} Y_i - \sum_{i=1}^{N} \bar{Y} = N\bar{Y} - N\bar{Y} = 0$$

2 Random Variable and Random Sample

Random Variable

The basic building block of statistics is the random variable (sometimes abbreviated "rv"). A random variable is the quantifier of our world; it associates a number with each event in the world. A classic example of this is the number of heads in a sequence of 3 coin tosses. The events are the possible outcomes (e.g., HHH, HTH, TTH, ...), and we define the variable X as the number of heads in the sequence such that X associates a real number with each event. Another example is the random variable H which represents the height of a person. Of course, a person is a complicated set of events (genetic inheritance, nutritional background, ...). The H variable measures or quantifies only one aspect of this complicated set of outcomes that make up the person.

Random Sample and IID

A random sample is a collection of independent and identically distributed ("iid") random variables. Let $\{X_1, X_2, \ldots, X_N\}$ be N iid rvs. The important idea to remember here is that a random sample describes a mechanism for obtaining the data. The actual data we analyze is a realization or draw from the random sample. Realized values are often denoted lowercase, as in $\{x_1, x_2, \ldots, x_N\}$. To be thorough with the notation, we could say $X_1 = x_1$ to indicate that the random variable X_1 took on the value x_1 in our sample.

For example, if we draw one person out of each of N rooms in the building according to a random sampling method and make measurements of their heights, we have constructed a random sample $\{h_1, h_2, \ldots, h_N\}$. These observations are independent in the sense that the height of the person drawn from room B301 is unrelated to the height of someone drawn from room B302.

Distribution

All random variables have a probability density or distribution. For our random sampling of heights from people in the business school, we can think of the distribution of heights in the population of all business school students. Unless certain classes or programs attract unusually short or tall students, we can make the assumption that each time we select someone from a classroom we are drawing from the same distribution of height.

3 Mean and Variance of Linear Combinations of Random Variables

Some useful facts for scalar a and rvs X and Y :

- $E[aX] = aE[X]$
- $\text{Var}(aX) = a^2 \text{Var}(X)$
- $\text{Var}(X) = E\left[(X - \mu_X)^2\right] = E\left[(X - \mu_X) X\right]$

For the bivariate case

- $E[aX + bY] = aE[X] + bE[Y]$
- $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \,\text{Cov}(X, Y)$

Extending to N-variates

- $E\left[\sum a_i Y_i\right] = \sum a_i E\left[Y_i\right]$
- $\text{Var}\left(\sum a_i Y_i\right) = \sum_i a_i^2 \text{Var}\left(Y_i\right) + 2 \sum_{i>j} a_i a_j \,\text{Cov}\left(Y_i, Y_j\right)$

Note that if the rvs are independent, the covariance terms are zero and the above expressions simplify. For example, under independence, $\text{Var}\left(\sum a_i Y_i\right) = \sum_i a_i^2 \text{Var}\left(Y_i\right)$.

4 Covariance and Correlation

The covariance between two random variables, say X and Y, is defined to be

$$\text{cov}(X, Y) = E\left[(X - \mu_X)\left(Y - \mu_Y\right)\right] = \sigma_{XY}$$

Covariance is a measure of linear dependence between X and Y. To estimate σ_{XY}, we use the sample covariance s_{XY}, defined to be

$$s_{XY} = \frac{1}{N-1} \sum_i (X_i - \bar{X})(Y_i - \bar{Y})$$

Interpretation of Covariance

Direction. Suppose the covariance is positive. This means that X and Y are both on the same side of the mean more frequently than on opposite sides, which suggests a positive linear relationship.

Magnitude. It is difficult to say what a *large* value of covariance is. For these reasons, the correlation coefficient was introduced. The correlation coefficient is the ratio of the covariance to the product of the standard deviations and it lies in the interval $[-1, 1]$:

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_Y}$$

Note that because standard deviations are always positive, the sign of ρ is determined by the sign of the covariance. The sample correlation coefficient is estimated by inserting the sample estimates of covariance and standard deviations:

$$r = \frac{s_{XY}}{s_X s_Y}$$

5 Normal Distribution

The normal distribution is the building block of all statistics. Nothing is ever exactly normally distributed, but the normal distribution can serve as a good approximation, especially in regression analysis.

There are many different normal distributions. The normal distribution has two parameters: the mean (μ) and the variance (σ^2, or equivalently the standard deviation σ). If you tell me μ and σ, then I know everything about this distribution. Thus, I only have to carry around two numbers to summarize the complete distribution rather than a large table like a histogram.

Two key features of the normal distribution:

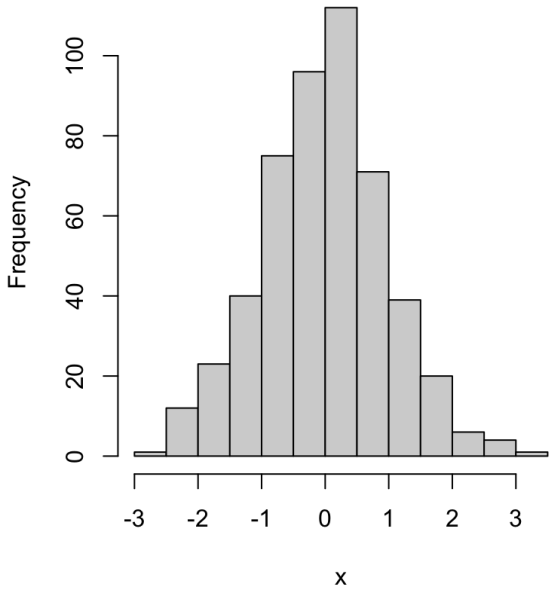
1. It's symmetric about the mean (μ = mode = median).
2. It has thin tails ($\Pr[|Y - \mu| > 3\sigma] = 0.0026$).

Let's look at some random samples from various normal distributions. The R commands to generate random draws from distributions begin with `r` and include `rbeta`, `rbinom`, `rcauchy`, `rchisq`, `rexp`, `rf`, `rgamma`, `rmultinom`, `rnorm`, `rpois`, `rt`, and `runif`, among others.

For example, we want to simulate a vector of 500 independent, standard (i.e., $\mu = 0$ and $\sigma = 1$) normal rvs. We use the `rnorm` function, which has defaults of $\mu = 0$ and $\sigma = 1$. Therfore, we can simply type

```
x <- rnorm(500)      # mean 0, sd 1
hist(x)
```

Notice that because we are randomly drawing only 500 realizations from these distributions, the histograms made from these relatively small samples display some sampling error. That is, they do not perfectly reflect the underlying data generating process.



Population Moments and Sample Moments

The population moments are the expected or average values taken with respect to the population distribution rather than sample distribution. Notably, The population moments are not necessarily the same as the sample moments. Specifically, if $Y_i \sim N(\mu, \sigma^2)$, then:

- Population Mean: $E[Y_i] = \mu$
- Sample Mean: $\bar{Y} = \frac{1}{N} \sum_i Y_i$

- Population Variance: $E[(Y_i - \mu)^2] = \sigma^2$
- Sample Variance: $s^2 = \frac{1}{N-1} \sum_i (Y_i - \bar{Y})^2$

- Population Std. Dev.: σ
- Sample Std. Dev.: s_Y

Z-Scores

If $Y \sim N(\mu, \sigma^2)$, then

$$Z = \frac{Y - \mu}{\sigma} \sim N(0, 1).$$

A $N(0, 1)$ random variable is often denoted Z. For example, the Central Limit Theorem tells us that, for large N, the mean will be normally distributed. So a z-score for the mean is

$$Z = \frac{\bar{Y} - \mu_0}{\sigma / \sqrt{N}} \quad \text{under} \quad H_0 : \mu = \mu_0$$